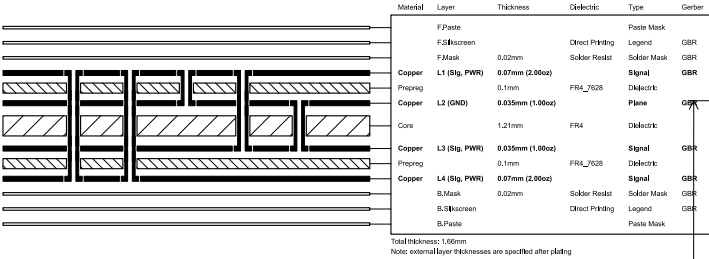
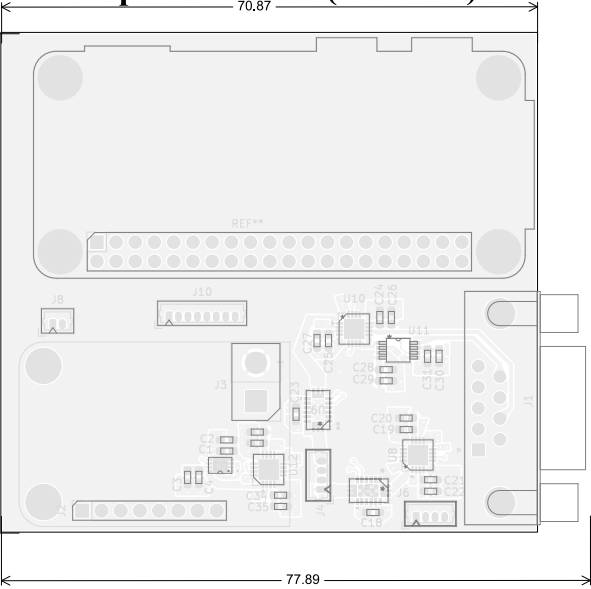


alb_mlb test Fabrication Document

Layer Stack Legend



Top Fabrication (Scale 1:1)



Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
B. Tg 170 C OR EQUIVALENT.
C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY GENERAL ATOMICS EMS.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE70.968 × 66.091 mm
BOARD THICKNESS1.660 mm
TRACE WIDTH0.200 mm
TRACE TO TRACE0.200 mm
MIN. HOLE (PTH)0.200 mm
MIN. HOLE (NPTH)2.750 mm
ANNULAR RING0.150 mm
COPPER TO HOLE0.254 mm
COPPER TO EDGE0.250 mm
HOLE TO HOLE0.254 mm
- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

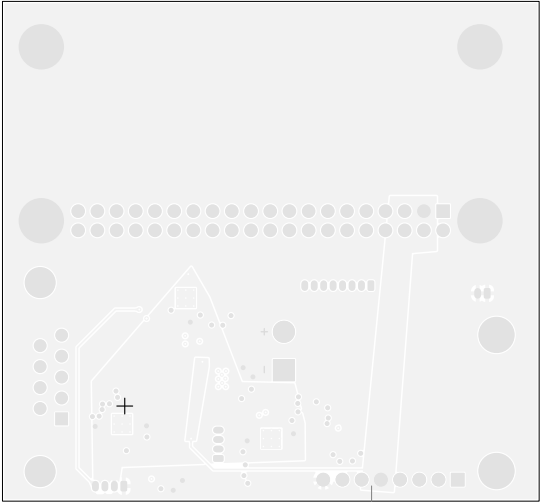
All dimensions are in millimeters unless otherwise specified.

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	Sheet Title: Top Fabrication (Scale 1:1)	Board Name: alb_mlb test		Project Name: ALB/MLB test	
	Sheet Path:	File Name: proalbmbl.kicad_pcb	Designer: Marcus Reighter	Date: 2024-04-13	Revision: + (Unreleased)
			Reviewer:	Size: A4	Sheet: 1 of 14

alb_mlb test Fabrication Document

3 × 120°

Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

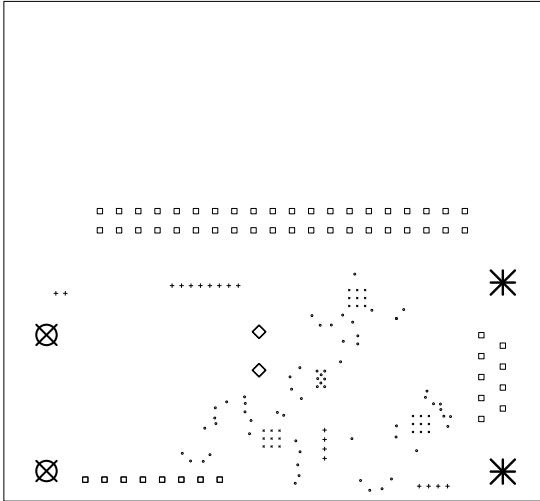
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	Sheet Path:		Reviewer:	Size: A4	Sheet: 2 of 14

alb_mlb test Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	27	0,20mm (7,87mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
○	63	0,25mm (9,84mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Via
+	18	0,50mm (19,69mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
□	65	1,00mm (39,37mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
◇	4	1,75mm (68,90mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
⊗	2	2,70mm (106,30mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
※	2	3,20mm (125,98mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
Total 181						

Drill Drawing L1 - L4 (Scale 1:1)



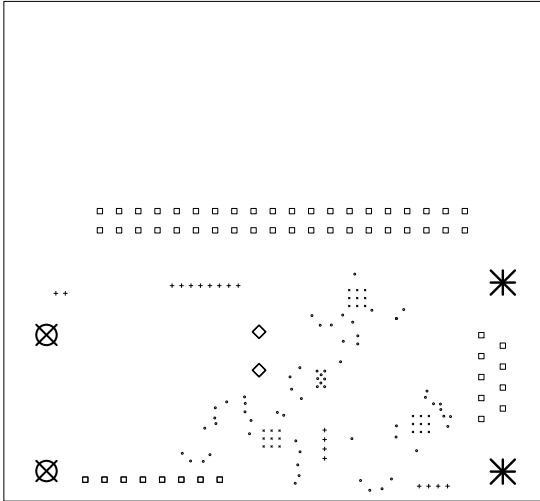
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		Board Name: alb_mlb test		Project Name: ALB/MLB test	
	Sheet Title: Drill Drawing (L1 - L4)	File Name: proalbmbl.kicad_pcb	Designer: Marcus Reighter	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 14

alb_mlb test Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	27	0,20mm (7,87mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
○	63	0,25mm (9,84mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Via
+	18	0,50mm (19,69mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
□	65	1,00mm (39,37mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
◇	4	1,75mm (68,90mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
⊗	2	2,70mm (106,30mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
⌘	2	3,20mm (125,98mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
Total 181						

Drill Drawing L1 - L4 (Scale 1:1)



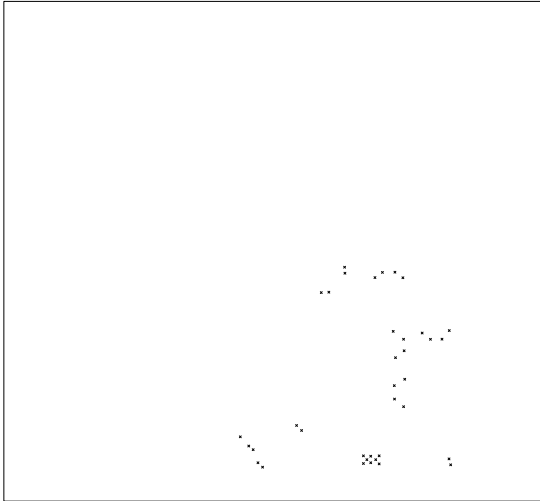
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	Sheet Path:		Reviewer:	Size: A4	Sheet: 4 of 14

alb_mlb test Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	37	0.25mm (9.84mils)	PTH	Round	L1 (Sig, PWR) - L2 (GND)	Via
	Total 37					

Drill Drawing L1 - L2 (Scale 1:1)



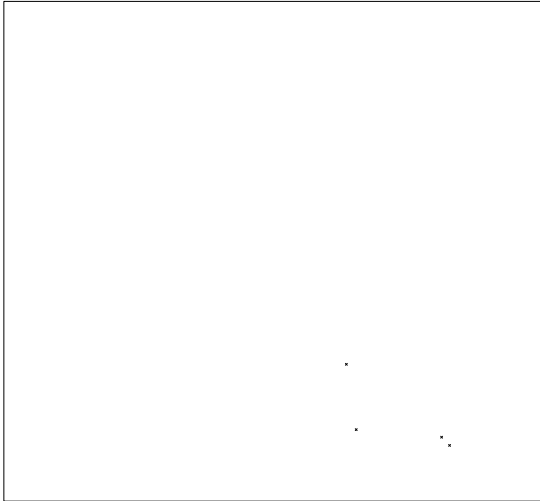
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	Sheet Path:		Reviewer:	Size: A4	Sheet: 5 of 14

alb_mlb test Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	0,25mm (9,84mils)	PTH	Round	L1 (Sig. PWR) - L3 (Sig. PWR)	Via
	Total 4					

Drill Drawing L1 - L3 (Scale 1:1)



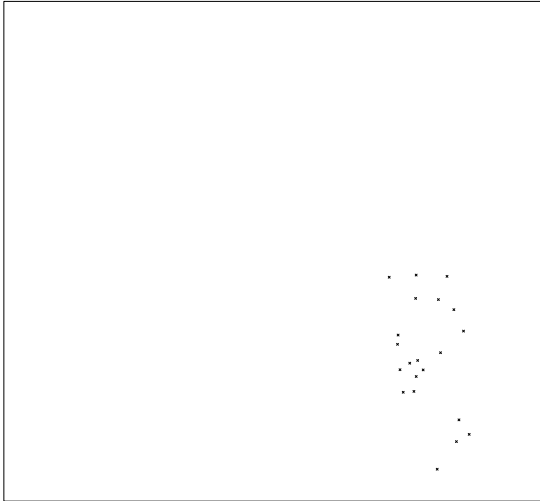
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	Sheet Path:		Reviewer:	Size: A4	Sheet: 6 of 14

alb_mlb test Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	21	0.25mm (9.84mils)	PTH	Round	L2 (GND) - L3 (Sig, PWR)	Via
	Total 21					

Drill Drawing L2 - L3 (Scale 1:1)



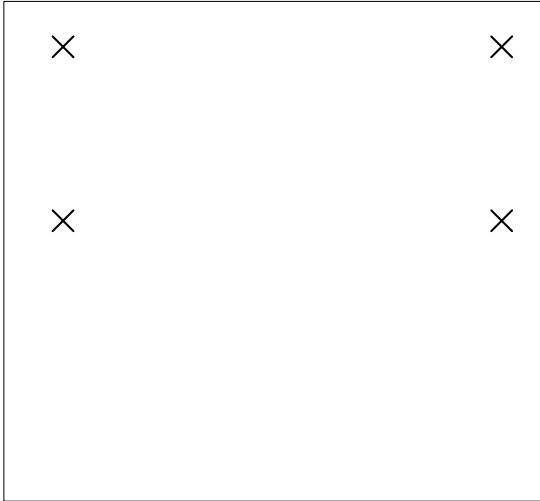
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	Sheet Path:		Reviewer:	Size: A4	Sheet: 7 of 14

alb_mlb test Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	2.75mm (108,27mils)	NPPTH	Round	L1 (Sig. PWR) - L4 (Sig. PWR)	Mechanical
	Total 4					

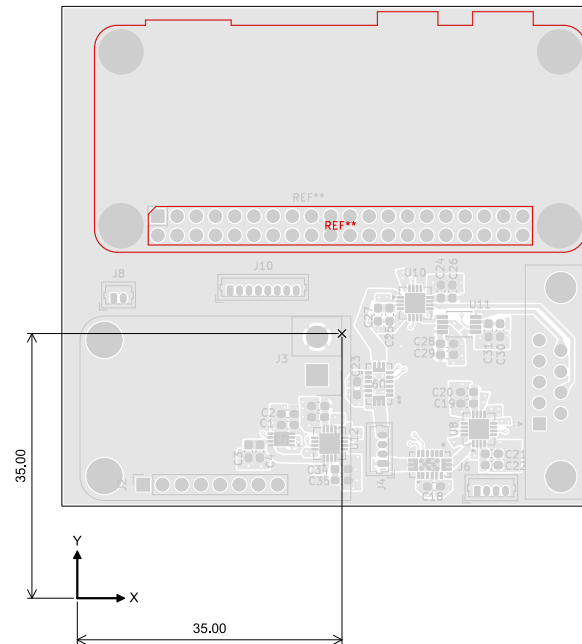
Drill Drawing L1 - L4 (Scale 1:1)



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	Sheet Path:		Reviewer:	Size: A4	Sheet: 8 of 14

alb_mlb test Fabrication Document

Top Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
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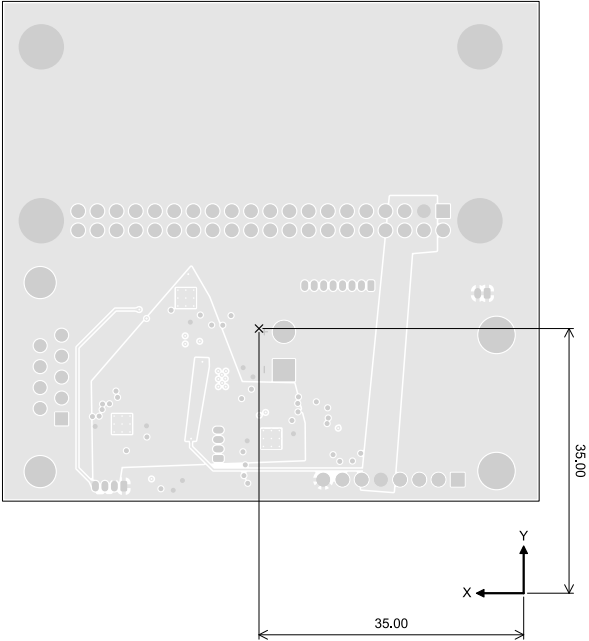
Ref.	Net	X [mm]	Y [mm]
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All dimensions are in millimeters unless otherwise specified.

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	Sheet Path:		Reviewer:	Size: A4	Sheet: 9 of 14	

alb_mlb test Fabrication Document

Bottom Test Points (Scale 1:1)



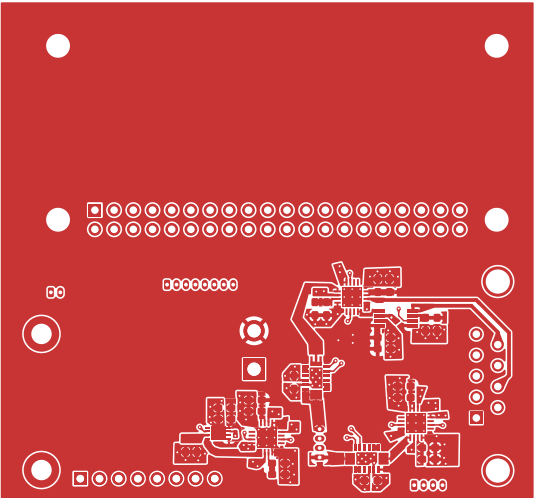
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All dimensions are in millimeters unless otherwise specified.

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alb_mlb test Fabrication Document

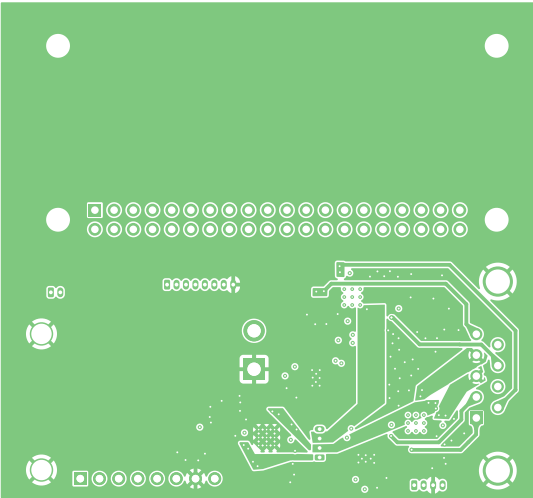
L1 (Sig, PWR) (Scale 1:1)



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	Sheet Title: L1 (Sig, PWR) (Scale 1:1)	File Name: proalbmlb.kicad_pcb	Designer: Marcus Reighter	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 11 of 14

alb_mlb test Fabrication Document

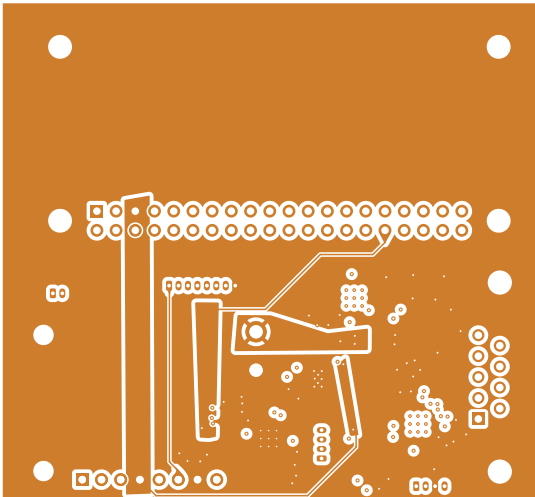
L2 (GND) (Scale 1:1)



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	Sheet Path:		Reviewer:	Size: A4	Sheet: 12 of 14

alb_mlb test Fabrication Document

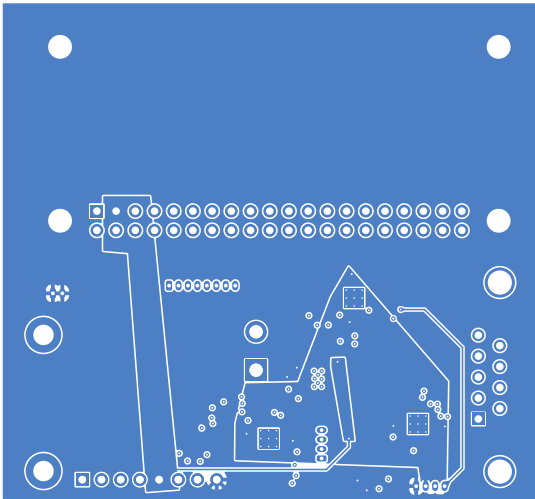
L3 (Sig, PWR) (Scale 1:1)



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	Sheet Title: L3 (Sig, PWR) (Scale 1:1)	File Name: proalbmlb.kicad_pcb	Designer: Marcus Reighter	Date: 2024-04-13	Revision: + (Unreleased)
	Sheet Path:		Reviewer:	Size: A4	Sheet: 13 of 14

alb_mlb test Fabrication Document

L4 (Sig, PWR) (Scale 1:1)



	Comments:	Company: General Atomics EMS		Variant: PRELIMINARY	Git Hash: bab92f0
		Board Name: alb_mlb test		Project Name: ALB/MLB test	
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